Public

TEST &
VALIDATION



CTAM: Three Years of Bridging Hyperscalers and Suppliers Through GPU Accelerator Compliance

[Basha Khader](#) *Firmware Engineer, AMD*

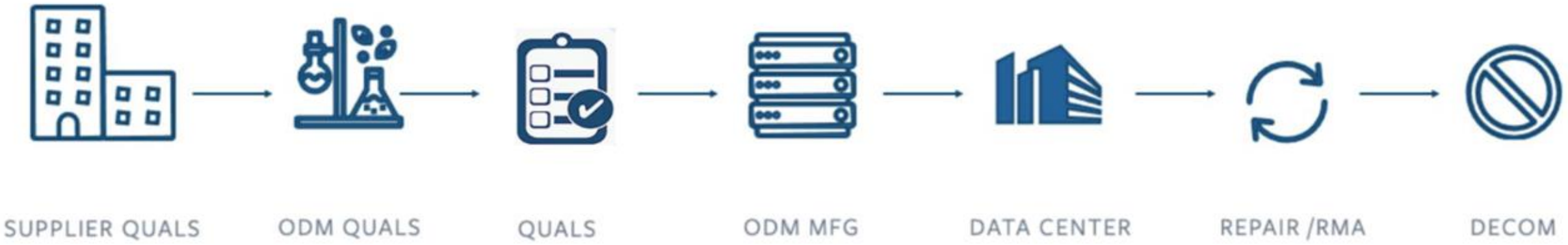
[Aditya Mishra](#) *Firmware Engineer, Nvidia*



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What is OCP Test and Validation (OCPTV)

Creating portable hardware diagnostics under the OCP Umbrella to reach a common goal.



Standardize Integration, Validation, Debug and Diagnostics at all stages of the Product Life Cycle

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The Problem Before CTAM

SUPPLIER

"Our firmware passes our tests."

?

HYPERSCALER

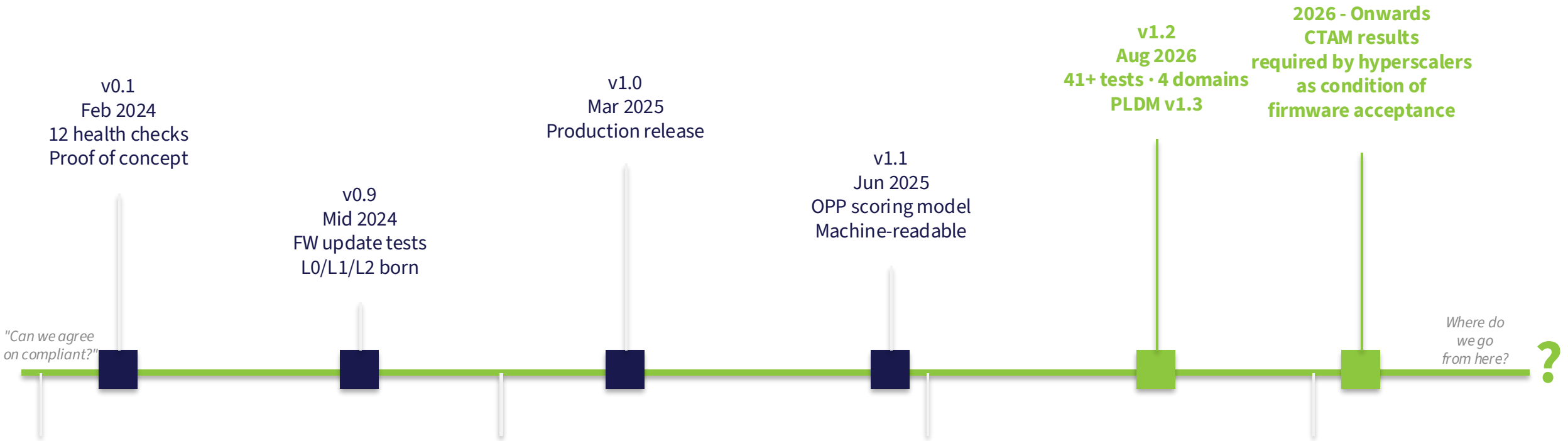
"Your firmware fails our lab."

No shared truth. · No common test. · No trusted number.

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The Journey

"From addressing nascent compliance needs to becoming the de-facto yardstick"



OCP Global Summit 2023

Introduction to the Industry
—
"Why CTAM compliance suite needs to be a standard?"

OCP Global Summit 2024

Submit feature requests —
CTAM becomes
community-driven

OCP Global Summit 2025

Industry adoption —
CTAM the de-facto measure
of firmware quality

OCP Global Summit 2026

Gold standard —
co-presenting as the
industry requirement

NVIDIA · AMD · Microsoft · OCP WG

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CTAM 2026 Evolution

Improved Compliance

Platform Maturity
Technical Foundation

- Spec 1.2 aware compliance
- Improved stability
- FW update enhancements
- PLDM v1.3 support

Supplier Adoption

Supplier Benefits
Ecosystem Enablement

- Early compliance validation
- OCP standards alignment
- Redfish interoperability
- Reduced integration risk

Hyperscaler Confidence

Hyperscaler Benefits
Deployment Confidence

- Common acceptance criteria
- OPP-driven decisions
- Risk-based reporting
- Faster Time to Market (TTM)

CTAM Platform

Smarter Testing → Earlier Compliance Detection → Faster Qualification → Greater Industry Trust

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What's Next

EOY 2026

Scenarios:

- PLDM v1.3 support + PLDM bundling compliance checks

June 2027

Features:

- CTAM support for mockups + redfish emulation models
- AI skills development for onboarding, scenario creation and debugging

Scenarios:

- New compliance for OCP spec levels 1.9/2.0
- Recovery image support

FAR

Vision:

- RAS API integration to cover GPU RAS compliance
- Extend CTAM scenarios for beyond AI nodes.

github.com/opencomputeproject/ocp-diag-ctam · Run it → File an issue → Submit a PR

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Call to Action

- Problem to solve
  - Enable hyperscalers and suppliers to focus on innovation by standardizing the foundational manageability framework. And then use CTAM to enforce the standards!
- How to get involved in the project/sub-project community
  - opencomputeproject/ocp-diag-ctam (github.com) (ocptv-ctam · PyPI)
  - Suggestions to invoke other open-source tools and integrate as a part of CTAM
- Where to find additional information (URL links)
  - OCP GPU FW Update Specification - https://www.opencompute.org/documents/ocp-gpu-fw-update-specification-v1-1-pdf
  - OCP GPU Manageability Interfaces Specification - https://www.opencompute.org/documents/ocp-gpu-accelerator-management-interfaces-v1-pdf
  - OCP GPU RAS Specification - https://www.opencompute.org/documents/ocp-gpu-and-accelerators-ras-requirements-v1-7-10-23-2025-pdf

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Thank You!

